Co-Wingman: A Hybrid Edge–Cloud AI Assistant for Real-Time Conversational Recovery Cues

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University of Florida | Master's in AI Systems | AI Days 2026

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Motivation

Help speakers recover a thought with a short, context-aware cue during a conversational pause.

Prototype capabilities

1. Local speech recognition
2. Rolling 60-second context
3. Manual or pause-triggered cues
4. Optional portfolio grounding
5. Minute summaries and S3 continuity

Privacy & control

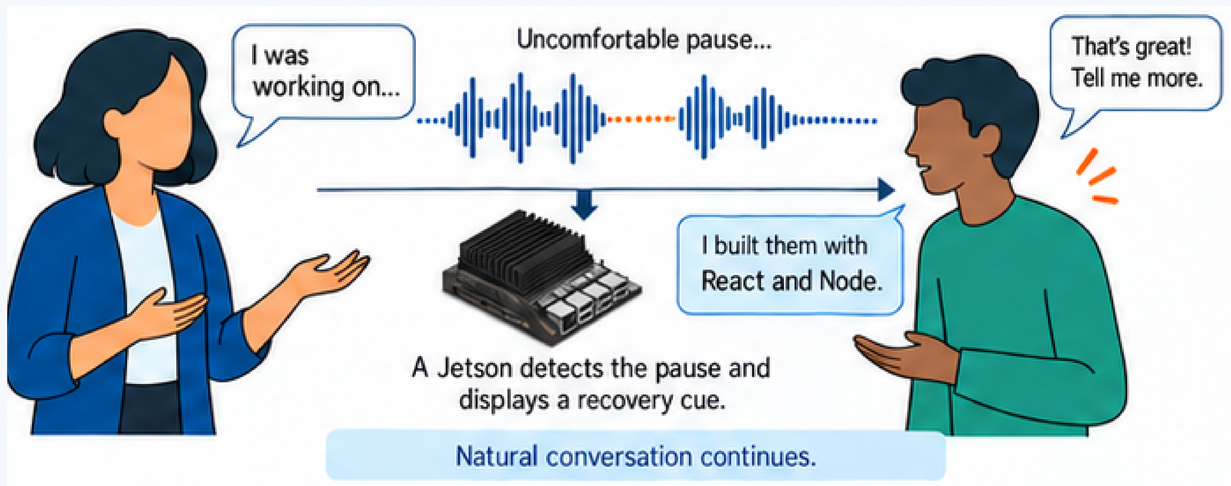
Raw audio stays on the Jetson.

Only bounded text reaches UF NaviGator.

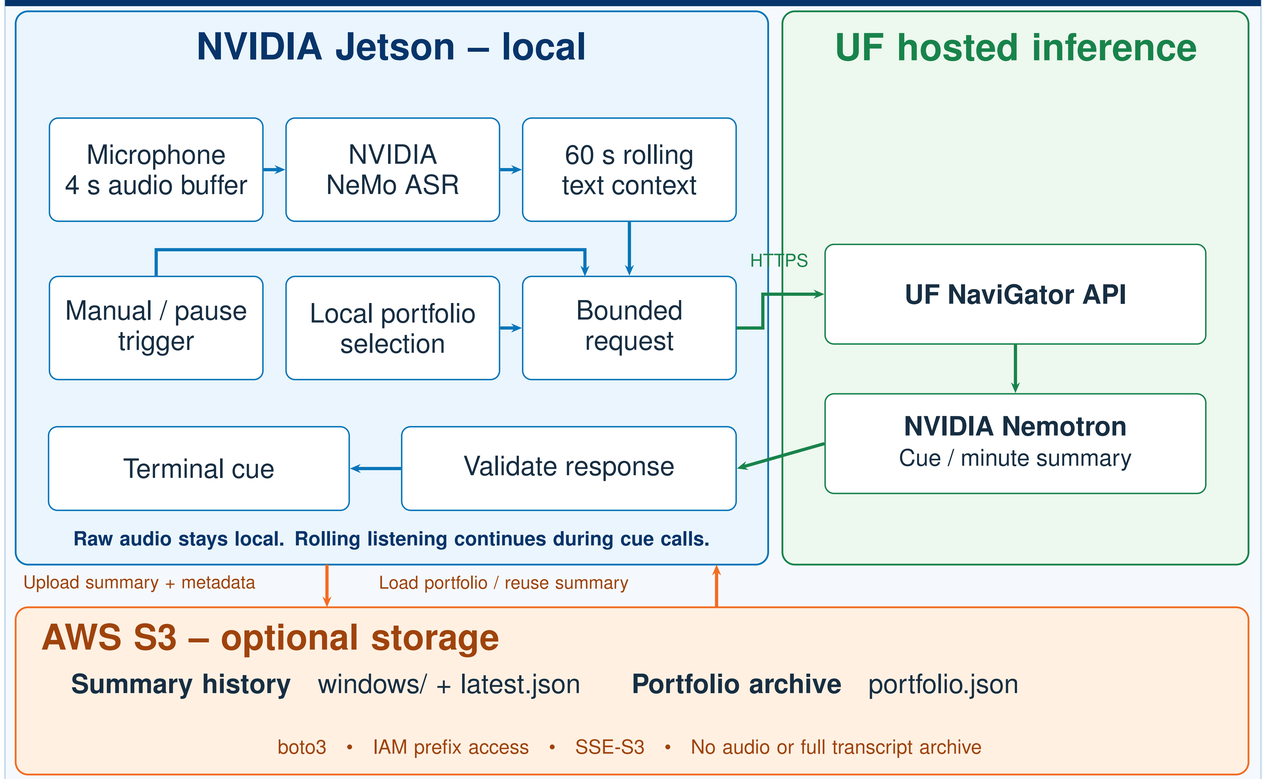
AWS storage is opt-in.

Pause assistance detects silence, not anxiety or speaker identity.

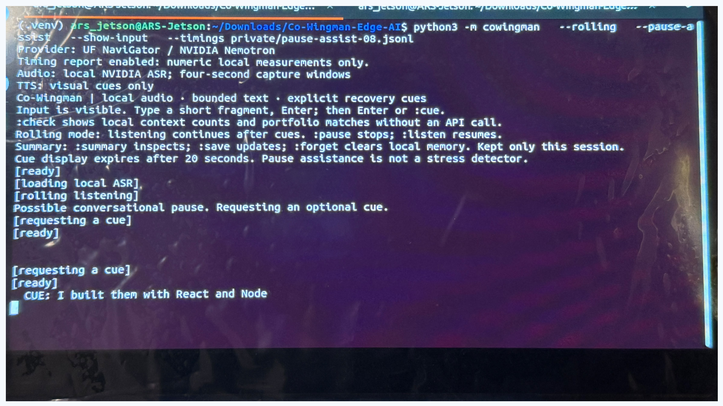
Speak → Pause → Recover



Hybrid architecture



Results on Jetson



Live visual cue demonstrated on Jetson

CUE: I built them with React and Node

94	13	0
audio windows	cue requests	dropped windows

Measured stage	Median	p95
ASR inference	110 ms	127 ms
Cue handling	1.04 s	2.54 s
Summary update	1.42 s	1.99 s

Timing sample: pause-assist-05; photo from a separate demo. 69 voiced windows, 7 summaries, zero recorded stage failures. Cue timing excludes capture wait; a completed request may offer no cue.

AWS integration implemented; live upload/reload validation pending. AWS was not included in this timing test.

Next steps

1. Reduce four-second buffering delay
2. Evaluate cue usefulness with users
3. Validate AWS upload and recovery
4. Evaluate experimental disfluency scoring

Local speech processing • Bounded cloud reasoning • Optional persistent memory

github.com/Co-Wingman-AI/Co-Wingman-Edge-AI